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RESEARCH ARTICLE



Business process digitisation and firm innovation performance: The role of knowledge search and digital culture

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ABSTRACT

Drawing on the knowledge-based view, this study examines the effect of business process digitisation on firm innovation performance. Furthermore, we investigate the mediating role of knowledge search (i.e., local search and boundary-spanning search) and the moderating effect of digital culture by focusing on the relationship between business process digitisation and firm innovation performance. The results, based on 330 questionnaire survey responses, show that business process digitisation has a positive effect on firm innovation performance, knowledge search plays a mediating role between business process digitisation and firm innovation performance, and digital culture has a positive moderating effect, which is mainly reflected in business process digitisation and boundary-spanning search. This study provides inspiration for firms to improve innovation performance through business process digitisation and expands the literature on knowledge search and firm innovation in the digital context.

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1. Introduction

With the advent of digital economy, digital technologies such as big data and artificial intelligence are continually reshaping the market competition logic and value chain (Hautala-Kankaanpää, 2022; Xia et al., 2022). Theorists and practitioners have widely regarded digital transformation as the key driver in seizing innovation opportunities and achieving value creation (Matarazzo et al., 2021; Vial, 2019), highlighting the critical influence of digital transformation on collaboration (Peng & Tao, 2022) and knowledge management (De Bem Machado et al., 2022) to promote innovation. Business process digitisation is a fundamental phase of digital transformation (Verhoef et al., 2021), which focuses on the application of digital technology to develop new business processes and improve existing processes (Adomako et al., 2021). Previous research has suggested that if organisations do not continuously apply digital technologies to improve and automate business processes, digital transformation will not achieve the desired results (Erceg & Zoranović, 2022; Heberle et al., 2017). In the process of digital transformation, business process digitisation directly affects the innovation processes of firms. It helps firms form explicit knowledge through the integration of key data and tacit knowledge of different business units (Xia et al., 2022) and enhances the relationship between firms and external stakeholders (Adomako et al., 2021), thereby promoting innovation. Based on the above discussion, exploring the effect of business process digitisation on innovation is conducive to deepening the

understanding of how digital transformation influences firm innovation.

Current research has showed that business process digitisation is associated with innovation strategy (Barnir et al., 2003) and innovation capacity (Ferreira et al., 2019). However, as Ferreira et al. (2019) argued, deeper and more detailed research on the implications of digital transformation for innovation, such as the impact of digital processes on innovation, has not received sufficient attention. To the best of the authors' knowledge, studies exploring the underlying processes of business process digitisation on innovation are scarce. In addition, although Barnir et al. (2003) explored the moderating impact of firm size and age on the link between Internet-enabled business process digitisation and innovation strategy, the boundary conditions of the effect of business process digitisation on firm innovation performance still need to be further explored.

To address these gaps, this study intends to explore the indirect effect of business process digitisation on firm innovation performance through knowledge search at first. According to the knowledge-based view, knowledge is a firm's most strategically important resource (Grant, 1996). Innovation is based on knowledge (Duke et al., 2022; Hassan & Raziq, 2019) and can be summarised as the reconfiguration of knowledge elements in essence (Roper et al., 2017; Yu et al., 2020). Through knowledge search, defined as the knowledge acquisition activity of firms to solve problems (Nelson & Winter, 1982), firms can realise the integration, reconfiguration and creation of

knowledge (Katila & Ahuja, 2002; Laursen & Salter, 2006), break the innovation bottleneck (Gao et al., 2020) and further facilitate the process of innovation (Yu et al., 2020). Business process digitisation reshapes business processes and integrates partners in the value chain into digital ecosystem (Li et al., 2020). It further broadens the channels for knowledge search (Lanzolla et al., 2021), and reframes the methods of knowledge search (De Bem Machado et al., 2022), thereby facilitating firms to grasp new opportunities and develop innovation activities. Previous research has mentioned that integrating digital technology into business process contributes to expanding the scope of knowledge search (Xia et al., 2022) and emphasised the importance of knowledge for innovation in the digital context (Arias-Pérez et al., 2021; De Bem Machado et al., 2022; Lanzolla et al., 2021). However, to the best of the authors' knowledge, few scholars have attempted to empirically examine the important role of knowledge search in the relationship between business process digitisation and firm innovation performance.

In addition, firms are more likely to exploit digital technology and create value if their own values match the values embedded within digital technologies (Martínez-Caro et al., 2020). And knowledge-related processes are often related to the social context (Duke et al., 2022). In the digital context, digital culture is defined as the beliefs and values within the organisation connected to the application of digital technology, and is related to a firm's acceptance of digital technology (Hautala-Kankaanpää, 2022). It can inspire employees to form digital vision, digital thinking, and subjective initiative (Wang et al., 2022). As a result, employees are more willing to apply digital technology to business processes (Martínez-Caro et al., 2020), which further stimulates knowledge sharing among employees and contributes to knowledge mining from big data. Correspondingly, the implementation effectiveness of knowledge search will also be improved (Wang et al., 2022), and then facilitating firm innovation. Although digital culture has been observed to influence the result of digitalisation (Hautala-Kankaanpää, 2022; Martínez-Caro et al., 2020), how it interacts with business process digitisation on knowledge search and firm innovation performance has not been empirically tested.

Accordingly, based on the knowledge-based view, this study conducts a questionnaire survey to investigate the impact of business process digitisation on firm innovation performance through knowledge search and moderating role of digital culture. In this way, we intend to make three contributions. Firstly, this study extends the literature on the effect of business process digitisation on innovation and deepens the understanding of innovation process in the context of digital transformation. Secondly, our study enriches the literature on knowledge management in

the digital context by exploring the mediating role of knowledge search. Thirdly, this study contributes to deepening the understanding of digital culture and enriching the literature on digital culture by revealing the contextual role of digital culture.

The remainder of this paper is organised as follows. In Section 2, we review the theoretical background and develop the hypotheses. In Section 3, we introduce the data collection, variable measurement, and data analysis. In Section 4, we present our results. Finally, in Section 5, we provide the discussion and conclusions.

2. Theoretical background and hypotheses

2.1. Business process digitisation and innovation performance

Digitisation is the process of applying digital technology. And digital technology is the combination of information, computing, communication, and connectivity technologies (Bharadwaj et al., 2013). As the fundamental phase of digital transformation (Verhoef et al., 2021), business process digitisation emphasises the application of digital technology to business processes, which aims to realise the digitisation of business processes through the improvement of material flow, information flow and capital flow (Adomako et al., 2021).

We argue that business process digitisation has a positive effect on firm innovation performance. Firstly, business process digitisation enables firms to achieve integrated management of business processes, which contributes to breaking the boundary among organisations and realising information transparency within firms (Nambisan, 2017). Correspondingly, firms are easier to identify potential risks in business process (Wamba et al., 2017), so as to formulate a risk response plan to reduce the risk and uncertainty in the innovation process and improve innovation performance (Jayakrishnan et al., 2018). Meanwhile, information transparency is conducive to obtaining more comprehensive data, thereby providing new innovation possibilities for organisations to realise sustainable innovation. Secondly, business process digitisation extends the supply chain through the digital management of procurement, marketing, etc. (Forman & Zeebroeck, 2019), which is conducive to identifying and connecting potential external innovation partners, thereby realising collaborative innovation (Hensen & Dong, 2020). Thirdly, business process digitisation simplifies the production and management process, which can further accelerates the process of product and technological innovation (Barnir et al., 2003). For example, firms can transfer the data generated from testing to the R&D and production process in real time by digitising business process, which is conducive to designing new products and

updating existing products rapidly, thereby improving firm innovation performance. Therefore, we propose the following hypothesis:

H1: Business process digitisation has a positive effect on innovation performance.

2.2. Business process digitisation and knowledge search

Business process digitisation promotes the flow and sharing of knowledge in business processes, which creates good conditions for knowledge search. According to whether the searched knowledge crosses organisational or technical boundaries, knowledge search can be divided into local search and boundary-spanning search (Hu et al., 2021; Yu et al., 2020). Local search can be defined as a knowledge search activity that takes place within existing geographical boundaries or technological base (Katila & Ahuja, 2002). And boundary-spanning search is a broader knowledge search activity that crosses organisational boundaries and knowledge base (Stuart & Podolny, 1996).

We argue that business process digitisation has a positive effect on both local search and boundary-spanning search. On the one hand, business process digitisation creates conditions for expanding storage space and improving the efficiency of data processing and exchange (Wu et al., 2019), which is conducive to realising knowledge selection, integration, and reconfiguration within the organisations (Martínez-Caro et al., 2020; Urbinati et al., 2020). Correspondingly, the efficiency and quality of local search will be improved. On the other hand, business process digitisation contributes to promoting boundary-spanning search activities by changing the way firms interact with the external actors (Scuotto et al., 2017). Digitisation of business processes, such as procurement and marketing enhances the connection with suppliers and customers, and improves trust with external stakeholders (Coronado-Medina et al., 2020), which facilitating communication and knowledge exchange (Papa et al., 2018), and enabling firms to acquire diversified external knowledge. Therefore, we propose the following hypotheses:

H2a: Business process digitisation has a positive effect on local search.

H2b: Business process digitisation has a positive effect on boundary-spanning search.

2.3. Knowledge search and innovation performance

Knowledge search is the knowledge acquisition activity of firms to solve problems (Nelson & Winter, 1982), which contributes to realising the integration, reconfiguration and creation of knowledge (Katila & Ahuja, 2002; Laursen & Salter, 2006) and further facilitates the process of innovation (Yu et al., 2020). Local search help firm achieve the deep mining of knowledge related to their technological base and promote the integration and reconfiguration of knowledge within organisations, thereby improving firm innovation performance (Hu et al., 2021; Stuart & Podolny, 1996). Meanwhile, boundary-spanning search is conducive to acquiring and integrating extensive and diversified knowledge from external actors. It further expands the existing knowledge base to relieve the dilemma of insufficient and highly homogeneous knowledge, thereby laying the foundation for firm innovation (Hu et al., 2021; Yu et al., 2020). Furthermore, boundary-spanning search contributes to improving the ability to predict technological and market changes, which making a direction on innovation, thereby improving firm innovation performance (Katila & Ahuja, 2002). Therefore, we propose the following hypotheses:

H3a: Local search has a positive effect on innovation performance.

H3b: Boundary-spanning search has a positive effect on innovation performance.

2.4. The mediating role of knowledge search

Existing research shows that the application of digital technology is an important driver for firms to acquire knowledge and further improve innovation performance. For example, Hensen and Dong (2020) found that the application of information technology in knowledge search enables firms to innovate more efficiently. Trantopoulos et al. (2017) suggested that information technology can improve process innovation performance by acquiring and integrating internal and external knowledge of the firm in the context of digital transformation. Business process digitisation facilitates the flow and reconfiguration of knowledge (Forman & Zeebroeck, 2019; Lewin et al., 2011). It enables firms to accumulate knowledge and deeply mine valuable knowledge through local search, which improves the efficiency of new product design and commercialisation (Hu et al., 2021), thereby improving innovation performance. Moreover, business

process digitisation expands the value chain by improving material flow, information flow and capital flow, and includes more suppliers and customers in process management. As a result, firms can identify and acquire external heterogeneous knowledge effectively (Hensen & Dong, 2020), thereby providing support for technology and product innovation (Martínez-Caro et al., 2020). Therefore, we propose the following hypotheses:

H4a: Local search mediates the relationship between business process digitisation and innovation performance.

H4b: Boundary-spanning search mediates the relationship between business process digitisation and innovation performance.

2.5. The moderating role of digital culture

Organisational culture is constituted of the values and beliefs shared by members of an organisation (Martins & Terblanche, 2003), which greatly influences the readiness and success of organisations in the adoption of knowledge management and digital transformation (Erceg & Zoranović, 2022). Therefore, the concept of digital culture has been derived in the context of digital transformation. Hautala-Kankaanpää (2022) defined digital culture as the beliefs and values within the organisation connected to the application of digital technology. Literature reveals that digital culture leads employees to form digital thinking and develop digital skills by sharing digital vision among organisation members (Martínez-Caro et al., 2020).

Knowledge search is an organisational learning behaviour (Mina et al., 2014) and organisational culture is an important environmental factor that influences the willingness of members to engage in learning activities (Bomm et al., 2022). In the context of digital transformation, we argue that digital culture plays a moderating role in the relationship between business process digitisation and knowledge search. On the one hand, the achievement of business process digitisation, such as digital platforms, enables the integration and sharing of knowledge in and among various departments (Xia et al., 2022). And digital culture relates to the acceptance of digital technologies (Hautala-Kankaanpää, 2022). With a higher level of digital culture, members in different departments are more willing to use digital platforms to dismantle barriers initiatively, and then search and transfer knowledge to achieve value co-creation (Bomm et al., 2022; Proksch et al., 2021; Wang et al., 2022), which realises a virtuous cycle of knowledge mining and creation, and further enhances the quality of local search. On the other hand, business process digitisation can achieve the

connectivity of systems and the integration of data between upstream and downstream firms (Coronado-Medina et al., 2020). And digital culture leads employees to regard digital technology as a source of value creation (Proksch et al., 2021). With a higher level of digital culture, employees are more inclined to enhance contacts with external stakeholders and dig deeper into customers' personalised needs through digital technologies, which is conducive to enhancing the breadth and depth of the boundary-spanning search. Therefore, we propose the following hypotheses:

H5a: Digital culture positively moderates the relationship between business process digitisation and local search.

H5b: Digital culture positively moderates the relationship between business process digitisation and boundary-spanning search.

Based on H4a, H4b, H5a, and H5b, we expect that digital culture positively moderates the indirect effect of business process digitisation on firm innovation performance through knowledge search. Specifically, with a higher level of digital culture, firms that implement business process digitisation possess more digital thinking. As a result, they prompt employees to strengthen the connection with internal and external partners through digital technology initiatively, and deeply explore the potential value of data, which contributes to enhancing the breadth and depth of the local search and boundary-spanning search, thereby improving innovation performance. However, with a lower level of digital culture, the willingness of employees to accept and apply digital technology diminishes. Therefore, they are less likely to initiatively search for knowledge through business process digitisation, making it difficult to obtain, integrate and utilise heterogeneous knowledge, and improve innovation performance. Therefore, we propose the following hypotheses:

H6a: Digital culture positively moderates the indirect effect of business process digitisation on innovation performance through local search.

H6b: Digital culture positively moderates the indirect effect of business process digitisation on innovation performance through boundary-spanning search.

Based on the research hypotheses, the theoretical model is shown in Figure 1.

3. Methodology

3.1. Sample and data collection

To test hypotheses, we conducted a survey using a structured questionnaire in China. And in order

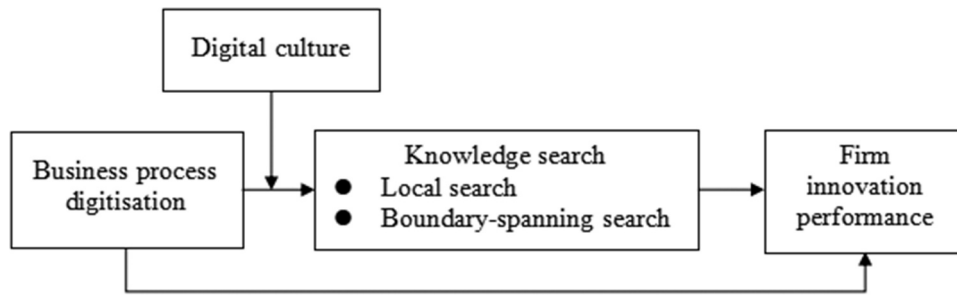


Figure 1. Theoretical framework.

to ensure the reliability and validity of the survey, we adapted the items from established instruments in previous literature to account for language differences and relevance to organisational practices in China at first. Secondly, we provided information on the first page of the questionnaire stating that this survey is only for academic purpose without any commercial purpose, and no firm or personal information will be disclosed. Finally, prior to conducting the actual questionnaire survey, 30 firms in Shanghai were selected to take a pre-research questionnaire.

To select the appropriate samples, we adopted four criteria. Firstly, the firms should be in good operating condition and have innovation output over a period of time. Secondly, considering the innovation cycle, the firms should be established for at least one year. Thirdly, due to the research purpose, the firms should have conducted business process digitisation and made some progress. Finally, in order to improve the universality of the conclusions, the firms in samples should be in a wide range of industries. Accordingly, we delivered 581 initial questionnaires through email and professional survey website “questionnaire star (www.wjx.cn)”. And the respondents were firm managers. Then, 428 questionnaires were received, with a sample return rate of 73.7%.

To ensure the authenticity of data, invalid questionnaires were eliminated in the following ways. Firstly, according to the number of questions and the speed of a normal person’s response, questionnaires with a response time of less than 150 seconds were eliminated. Secondly, invalid questionnaires were eliminated according to the selection questions set in the questionnaire. For example, business process digitisation emphasises the digitisation of procurement, production, management, and marketing, etc. Please select “does not match at all”. The questionnaires in which respondents did not choose “does not match at all” were eliminated. Thirdly, questionnaires with a high repetition rate and a regular response pattern were eliminated. Finally, a total of 98 invalid questionnaires were excluded, and 330 questionnaires were retained, which constituted a response rate of 56.8%.

Out of a total sample of 330, 15.2% of firms had less than 100 employees, 42.4% were firms with 101 to 500 employees, 24.5% of firms had 501–1000 employees, and 17.9% were from firms with more than 1000 employees. In terms of firm age, 5.8% of firms were aged less than 5 years, 49.7% of firms were from the age group of 6 to 10 years, 25.4% of firms belonged to the age group of 11 to 25, and 19.1% of firms were older than 25 years. In terms of firm property, 36.4% of firms were state-owned, 44.5% were private firms, 10.3% were joint ventures, 8.2% were wholly foreign-owned, and 0.6% were “other”. Moreover, regarding to education levels, 3.0% of respondents had a high school degree or less, 26.4% had a junior college degree, 64.2% had a Bachelor-level degree, and 6.4% had a Master degree or above. And finally, regarding to industry, most firms were in the manufacturing industry (29.1%), followed by information transmission, software, and information technology services (21.2%), construction (10.3%), accommodation and food service activities (7.9%), financial activities (7.9%), the biological industry (7.9%), wholesale and retail trade (6.4%), other service activities (4.2%), real estate activities (3.3%) and education (1.8%). These characteristics are shown in Table 1.

3.2. Measures

All items were adapted from the established instruments in previous literature, and we used 7-point Likert scale to measure all these items, with “1” representing “strongly disagree” to “7” representing “strongly agree”.

Independent variable: business process digitisation. Drawing on Adomako et al. (2021), business process digitisation was identified by five items: “In our firm, we use digital technology to support the supplier relationships business”, “In our firm, we use digital technology to support the products and services business”, “In our firm, we use digital technology to support the production and operations business”, “In our firm, we use digital technology to support the marketing business”, and “In our firm, we use digital technology to

Table 1. Characteristics of the firms in the sample.

Dimension	Category	Number	Percentage (%)
Firm size	<100	50	15.2
	101–500	140	42.4
	501–1000	81	24.5
	>1000	59	17.9
Firm age	≤5	19	5.8
	6–10	164	49.7
	11–25	84	25.4
	>25	63	19.1
Firm property	State-owned	120	36.4
	Private	147	44.5
	Joint venture	34	10.3
	Wholly foreign-owned	27	8.2
	Others	2	0.6
Education	High school degree or less	10	3.0
	Junior college degree	87	26.4
	Bachelor-level degree	212	64.2
	Master degree or above	21	6.4
Industry	Wholesale and retail trade	21	6.4
	Manufacturing	96	29.1
	Construction	34	10.3
	Accommodation and food service activities	26	7.9
	Financial activities	26	7.9
	Information transmission, software and information technology services	70	21.2
	Real estate activities	11	3.3
	Education	6	1.8
	Biological industry	26	7.9
	Other service activities	14	4.2

support the customer relationships business”. The Cronbach’s alpha (α) reliability coefficient of the scale was 0.801.

Mediating variables: local search, boundary-spanning search. Drawing on Hu et al. (2021), Laursen and Salter (2006), Rosenkopf and Nerkar (2001), and Yu et al. (2020), local search was measured by five items: “In our firm, we innovate the key technology in the department”, “In our firm, we search the knowledge base to find and select knowledge”, “In our firm, we tap into existing areas of technology”, “In our firm, we identify and update the existing knowledge of the firm with the continuous development of new technology”, “In our firm, we search for knowledge that is close to an existing technical field”. In this study, the Cronbach’s alpha(α) reliability coefficient of the scale was 0.747. Moreover, drawing on Hu et al. (2021), Laursen and Salter (2006), and Yu et al. (2020), boundary-spanning search was measured by five items: “In our firm, we search for knowledge disclosed from patent gazette”, “In our firm, we search for knowledge from technology licensing”, “In our firm, we search for knowledge from research institutions, universities or colleges”, “In our firm, we search for knowledge from industry exhibitions”, “In our firm, we search for knowledge from technical publications and technical conferences”. The Cronbach’s alpha(α) reliability coefficient of the scale was 0.760.

Moderator variable: digital culture. Drawing on Martínez-Caro et al. (2020), digital culture was measured by five items: “In our firm, the teams conduct

functional collaboration in the process of innovation and digital transformation”, “In our firm, we have a clear direction for digital transformation in the firm culture”, “In our firm, the culture of digital transformation is a natural process”, “In our firm, we share the digital strategy with employees and consider the opinions and suggestions from employees”. The Cronbach’s alpha(α) reliability coefficient of the scale was 0.750.

Dependent variable: firm innovation performance. Drawing on previous studies (Chen et al., 2011; Zhang & Li, 2010), firm innovation performance was identified by six items: “Compared with competitors in the field, the number of new products in our firm is more”, “Compared with competitors in the field, the number of patent applications in our firm is more”, “Compared with competitors in the field, the proportion of sales revenue of new products in total sales is higher”, “Compared with competitors in the field, the speed of new product development is faster”, “Compared with competitors in the field, the success rate of innovative products in our firm is higher”, “Compared with competitors in the field, the market share of new products in our firm is higher”. The Cronbach’s alpha(α) reliability coefficient of the scale was 0.828.

Control variables: firm size, firm age, firm property, industry, and education, which may have a certain influence on the results.

3.3. Data analysis

3.3.1. Common method bias

We conduct common method variance by Harman’s single-factor test and common latent factor. EFA showed that the first factor without rotation explained only 35.669% of the total variance (less than 40%), which did not account for the majority of the variance. Then, the result of CFA indicated the change of fit indices: $\Delta\chi^2/df = 0.026$, $\Delta CFI = 0.002$, $\Delta TLI = 0.002$, $\Delta RMSEA = 0.002$. Based on these results, common method bias had no substantial influence on our results.

3.3.2. Reliability and validity assessment

In this study, Cronbach’s α and CR were used to examine constructs’ reliability. KMO and AVE were used to examine constructs’ validity, as shown in Table 2. The results showed that the Cronbach’s α for each construct was greater than 0.7, ranging from 0.747 to 0.828. Hence, internal consistency was high. And all factor loadings were significant and greater than 0.50. The estimates for CR were higher than 0.7, indicating that combined reliability was great. The estimates for KMO were greater than 0.6, ranging from 0.773 to 0.874, and the estimates for AVE were

greater than 0.5, indicating that the convergent validity was great.

We assessed discriminant validity of the constructs by conducting CFA. The results are shown in Table 3. The fit index of the five-factor model was significantly better than others, indicating that the five variables involved in this study had good discriminant validity and were indeed five different constructs.

Moreover, Table 4 presents the means, standard deviations, and variable correlations. The means and standard deviations were within a reasonable range. The correlation coefficients between constructs showed no abnormality and there were correlations between constructs. Thus, the hypotheses of direct effect were supported preliminarily. In order to check for multicollinearity, the variance inflation factor (VIF) was calculated. The results showed that the highest inflation factor was 1.994, which was much lower than the critical value of 10, indicating that there was no serious multicollinearity.

4. Results

To test the hypotheses, we used SPSS 22.0 for regression analysis, and the results are shown in Table 5. In Model 5, we tested the effects of control variables on firm innovation performance. Model 6 included business process digitisation on the basis of Model 5, and the results showed that business process digitisation had a significant positive effect on firm innovation performance ($\beta = 0.488, p < 0.001$). Thus, H1 was supported.

We followed Baron and Kenny's (1986) classic approach to mediation testing. The results are shown in Table 5. In our analyses, business process digitisation was positively associated with innovation performance

($\beta = 0.488, p < 0.001$). Moreover, the results in Model 2 showed that the effect between business process digitisation and local search was significant and positive ($\beta = 0.689, p < 0.001$), indicating that H2a was supported. The results in Model 7 showed that local search had a significant positive effect on firm innovation performance ($\beta = 0.539, p < 0.001$), indicating that H3a was supported. The results in Model 8 showed that after entering local search, the effect of business process digitisation on firm innovation performance was reduced ($\beta = 0.224, p < 0.001$). These findings were consistent with the conditions for partial mediation. Thus, H4a was supported. In addition, the results in Model 4 showed that business process digitisation had a significant positive effect on boundary-spanning search ($\beta = 0.513, p < 0.001$), indicating that H2b was supported. The results in Model 9 showed that boundary-spanning search had a significant positive effect on firm innovation performance ($\beta = 0.652, p < 0.001$), indicating that H3b was supported. And the results in Model 10 showed that after entering boundary-spanning search, the effect of business process digitisation on firm innovation performance was reduced ($\beta = 0.209, p < 0.001$). Again, these findings are in line with the conditions for partial mediation. Therefore, H4b was supported.

Moreover, we tested the mediating effects of local search and boundary-spanning search by the bootstrapping application with 5000 resamples. The results in Table 6 showed that the indirect effect of local search was significant ($\beta = 0.258$, CI ranging from 0.103 to 0.431, excluding 0), and the indirect effect of boundary-spanning search was significant ($\beta = 0.273$, CI ranging from 0.190 to 0.360, excluding 0). Thus, H4a and H4b were supported.

Table 2. Results of the reliability and validity assessment.

Constructs	Items	loadings	Cronbach's α	CR	AVE	KMO
Business process digitisation	BD1	0.751	0.801	0.863	0.557	0.831
	BD2	0.759				
	BD3	0.751				
	BD4	0.744				
	BD5	0.725				
Local search	LS1	0.671	0.747	0.832	0.500	0.796
	LS2	0.688				
	LS3	0.705				
	LS4	0.731				
	LS5	0.732				
Boundary-spanning search	BS1	0.719	0.760	0.840	0.513	0.802
	BS2	0.729				
	BS3	0.667				
	BS4	0.713				
	BS5	0.750				
Digital culture	DC1	0.766	0.750	0.843	0.572	0.773
	DC2	0.740				
	DC3	0.747				
	DC4	0.773				
Firm innovation performance	IP1	0.730	0.828	0.876	0.540	0.874
	IP2	0.717				
	IP3	0.745				
	IP4	0.706				
	IP5	0.754				
	IP6	0.756				

Table 3. Results of confirmatory factor analysis.

Model	χ^2	df	χ^2/df	CFI	TLI	RMSEA	SRMR
Five factor model(FIP, BPD,LS,BS,DC)	352.525	265	1.330	0.970	0.966	0.032	0.035
Four factor model(FIP+BPD,LS,BS, DC)	619.340	269	2.302	0.878	0.864	0.063	0.057
Three factor model(FIP+BPD,LS+BS,DC)	658.055	272	2.419	0.866	0.852	0.066	0.059
Two factor model(FIP+BPD,LS+BS+DC)	688.921	274	2.514	0.856	0.842	0.068	0.060
One factor model(FIP+BPD+LS+BS+DC)	720.788	275	2.621	0.845	0.831	0.070	0.061

Notes: "+" means combining two factors into one factor.

FIP = firm innovation performance; BPD = business process digitisation; LS = local search; BS = boundary-spanning search; DC = digital culture.

Table 4. Mean, standard deviation, and correlation coefficient.

Constructs	1	2	3	4	5	6	7	8	9	10
1.Firm size	1									
2.Firm age	0.248***	1								
3.Firm property	0.023	-0.040	1							
4.Industry	0.014	0.020	-0.140*	1						
5.Education	-0.001	-0.045	0.044	0.131*	1					
6.Business process digitisation	-0.038	-0.060	0.054	0.021	0.055	1				
7. Local search	0.031	0.045	0.033	0.042	0.134*	0.688***	1			
8.Boundary-spanning search	-0.027	0.046	0.023	0.084	0.113*	0.515***	0.619***	1		
9.Digital culture	0.012	-0.002	-0.015	0.011	0.088	0.659***	0.650***	0.574***	1	
10.Innovation performance	0.063	0.060	0.008	0.083	0.030	0.482***	0.538***	0.648***	0.494***	1
Means	2.452	2.821	1.921	4.448	2.739	5.654	5.747	5.618	5.662	5.547
S.D.	0.955	0.804	0.919	2.583	0.618	0.784	0.686	0.747	0.769	0.767

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

N = 330.

The results of moderating effects are shown in Table 7. First, in order to avoid multicollinearity, we centralised variables of business process digitisation and digital culture and then constructed interaction terms. Next, we made local search and boundary-spanning search as dependent variables. And then control variables, independent variable, moderator variable, the interaction terms of independent variable and moderator variable were included in the model successively. The results in Model 12 showed that the interaction item of business process digitisation and digital culture had no significant effect on local search ($\beta = 0.053$, $p > 0.05$). Thus, H5a was not supported. According to Model 14, the interaction term of business process digitisation and digital culture had a significant positive effect on boundary-spanning search ($\beta = 0.164$, $p < 0.01$). Therefore, H5b was supported. This is possibly because business process digitisation realises the digitisation of the business

process of procurement, production, and marketing, which can strength the connections with suppliers, customers, etc. And digital culture mainly improves employees' attention to the value of data (Proksch et al., 2021), promoting employees to mine valuable knowledge from external stakeholders through business process digitisation. However, local search is more likely to occur within the firm. Therefore, with a high level of digital culture, firms may conduct more boundary-spanning search through business process digitisation.

We further paid attention to the effect of business process digitisation on firm innovation performance through boundary-spanning search with different levels of digital culture. And bootstrapping was performed by PROCESS programme in SPSS 22.0. As shown in Table 8, the indirect effect of business process digitisation on firm innovation performance through boundary-spanning search was significant

Table 5. Regression results of business process digitisation, knowledge search and firm innovation performance.

Variables	LS		BS		FIP					
	Model 1	Model 2	Model 3	Model 4	Model5	Model 6	Model 7	Model 8	Model 9	Model 10
FS	0.018	0.037	-0.043	-0.030	0.050	0.063	0.040	0.049	0.078	0.079
FA	0.047	0.081*	0.061	0.086	0.047	0.072	0.022	0.041	0.008	0.025
FP	0.033	-0.005	0.033	0.005	0.019	-0.007	0.002	-0.005	-0.002	-0.010
IS	0.028	0.012	0.075	0.063	0.081	0.070	0.066	0.065	0.033	0.036
ED	0.131*	0.098*	0.104	0.080	0.021	-0.002	-0.050	-0.040	-0.047	-0.046
BPD		0.689***		0.513***		0.488***		0.224***		0.209***
LS							0.539***	0.383***		
BS									0.652***	0.543***
R ²	0.022	0.492	0.023	0.283	0.013	0.249	0.298	0.324	0.429	0.461
ΔR^2	0.007	0.483	0.008	0.270	-0.002	0.235	0.285	0.309	0.418	0.449
F	1.486	52.209***	1.509	21.294***	0.886	17.876***	22.840***	22.020***	40.417***	39.286***

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

N = 330; FS = firm size; FA = firm age; FP = firm property; IS = industry; ED = education; FIP = firm innovation performance; BPD = business process digitisation; LS = local search; BS = boundary-spanning search; DC = digital culture.

Standardised coefficients are shown.

when digital culture was high (95% CI = $[-0.006, 0.203]$, including 0), but was insignificant when digital culture was lower (95% CI = $[0.116, 0.374]$, excluding 0). According to Preacher et al. (2007), one indirect effect is significant and the other is not when the moderator takes different values, which indicates that the moderated mediation effect is supported. Thus, H6b was supported.

5. Discussion

5.1. Conclusions

Based on the knowledge-based view, this study explored the influence mechanism of business process digitisation on firm innovation performance by introducing knowledge search as mediator and analysed the moderating effect of digital culture. The conclusions of this study are as follows.

Firstly, business process digitisation has a positive effect on firm innovation performance. It provides evidence that firms which pay more attention to business process digitisation tend to achieve a higher level of innovation performance. Business process digitisation is conducive to realising integrated business management, which can accelerate the speed of product updates and reduce innovation risks, thereby improving innovation performance.

Secondly, local search and boundary-spanning search mediate the relationship between business process digitisation and innovation performance. According to the knowledge-based view, business

process digitisation provides opportunities and channels for interaction (Adomako et al., 2021), which creates conditions for local search and boundary-spanning search. And the further integration and utilisation of knowledge promotes the improvement of innovation performance (Roper et al., 2017; Yu et al., 2020).

Thirdly, digital culture positively moderates the relationship between business process digitisation and boundary-spanning search, and it also positively moderates the indirect effect of business process digitisation on innovation performance through boundary-spanning search. With a higher level of digital culture, employees regard digital technology as a source of value creation (Proksch et al., 2021). As a result, employees are more willing to cooperate with external stakeholders and deeply mining valuable knowledge with the application of digital technology, thereby enhancing the breadth and depth of knowledge search to improve firm performance.

5.2. Theoretical contributions

This study makes several important theoretical contributions. Firstly, we shed new light on the black box of how business process digitisation influences firm innovation performance and explore the boundary conditions. Prior studies ignored the mechanisms by which business process digitisation influences firms innovation performance. And the boundary conditions need to be further explored. Thus, this study extends the literature on the relationship between

Table 6. Results of bootstrap mediation effect.

Mediated path	Path coefficient	Standard error	95% confidence interval	
			Lower	Upper
BPD → LS → FIP	0.258	0.083	0.103	0.431
BPD → BS → FIP	0.273	0.044	0.190	0.360

Notes: FIP = firm innovation performance; BPD = business process digitisation; LS = local search; BS = boundary-spanning search.

Table 7. Moderating effect of digital culture between business process digitisation and knowledge search.

Variables	LS		BS	
	Model 11	Model 12	Model 13	Model 14
FS	0.018	0.028	−0.043	−0.037
FA	0.047	0.068	0.061	0.067
FP	0.033	0.014	0.033	0.028
IS	0.028	0.020	0.075	0.074
ED	0.131*	0.076*	0.104	0.049
BPD		0.486***		0.300***
DC		0.352***		0.460***
BPD*DC		0.053		0.164**
R ²	0.022	0.557	0.023	0.395
ΔR ²	0.007	0.546	0.008	0.380
F	1.486	50.455***	1.509	26.164***

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

N = 330; FS = firm size; FA = firm age; FP = firm property; IS = industry; ED = education; BPD = business process digitisation; LS = local search; BS = boundary-spanning search; DC = digital culture.

Standardised coefficients are shown.

Table 8. Results of moderated mediating effect.

Path	Condition of digital culture	Indirect effect	SE	95%CI	
				Lower level	Upper level
BPD→BS→FIP	Low	0.102	0.053	−0.006	0.203
	Middle	0.159	0.047	0.083	0.270
	High	0.216	0.063	0.116	0.374

Notes: FIP = firm innovation performance; BPD = business process digitisation; BS = boundary-spanning search.

business process digitisation and innovation. And it further deepens the understanding of innovation process in the context of digital transformation.

Secondly, we theorise firm innovation process based on the knowledge-based view and identify two critical channels through which business process digitisation can influence firm innovation performance from knowledge search perspective. In the digital context, prior studies have paid attention to the impact of knowledge on innovation and the role of knowledge search (Arias-Pérez et al., 2021; De Bem Machado et al., 2022; Lanzolla et al., 2021; Xia et al., 2022). However, as Lanzolla et al. (2021) suggested, how digitalisation and knowledge search filter down to innovation outcomes is less obvious. Thus, this study makes up for this shortcoming to some extent. In addition, our research answers the call for the research topic of knowledge management and digital transformation (De Bem Machado et al., 2022) to enrich the literature on knowledge management in the digital context.

Thirdly, we explore the contextual role of digital culture. Thus, this study answers Lanzolla et al.'s (2021) call to investigate the boundary conditions of when digitisation might enable different types of knowledge search. In addition, to the best of the authors' knowledge, this is the first study examining the moderating role of digital culture in the association between business process digitisation and innovation, which enriches the literature on the digital culture.

5.3. Practical contributions

The practical contributions of this research are as follows:

Firstly, with the advent of digital economy, implementing business process digitisation is an effective strategy to improve innovation performance. Managers can increase investment in business process digitisation, such as establishing digital platform, to realise the integration of data in business processes and connection with external stakeholders, which contributes to promoting firm innovation.

Secondly, managers should pay attention to the benefit of business process digitisation on knowledge search. Firms implementing business process

digitisation can consider establishing knowledge base to realise a virtuous cycle of knowledge mining and creation, and applying digital technology to mine valuable knowledge, which is conducive to the improvement of firm innovation performance.

Thirdly, when implementing business process digitisation to promote innovation, managers should make an effort to cultivate digital culture. Managers should transmitting the value of data and take some ways to encourage employees to utilise digital technology, such as conducting training and providing rewards.

5.4. Limitations and future research

This study has some limitations. Firstly, this study focuses on the effect of business process digitisation on firm innovation performance. In the future, the effect of other areas of digitisation, such as digital products and digital services can be further explored. Secondly, this study divides knowledge search into local search and boundary-spanning search. In the future, research could explore other categories of knowledge search in the relationship between business process digitisation and firm innovation performance, such as knowledge search depth and breadth, etc. Thirdly, this study uses cross-sectional data, and future research could track firm data and analyse panel data to increase the universality of the conclusions.

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References

- Adomako, S., Amankwah-Amoah, J., Tarba, S. Y., & Khan, Z. (2021). Perceived corruption, business process digitization, and SMEs' degree of internationalization in sub-Saharan Africa. *Journal of Business Research*, 123, 196–207. <https://doi.org/10.1016/j.jbusres.2020.09.065>
- Arias-Pérez, J., Velez-Ocampo, J., & Cepeda-Cardona, J. (2021). Strategic orientation toward digitalization to improve innovation capability: Why knowledge acquisition and exploitation through external embeddedness matter. *Journal of Knowledge Management*, 25(5), 1319–1335. <https://doi.org/10.1108/JKM-03-2020-0231>
- Barnir, A., Gallagher, J. M., & Auger, P. (2003). Business process digitization, strategy, and the impact of firm age and size: The case of the magazine publishing industry. *Journal of Business Venturing*, 18(6), 789–814. [https://doi.org/10.1016/s0883-9026\(03\)00030-2](https://doi.org/10.1016/s0883-9026(03)00030-2)
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variables distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), 1173–1182. <https://doi.org/10.1037/0022-3514.51.6.1173>
- Bharadwaj, A., El Sawy, O. A., Pavlou, P. A., & Venkatraman, N. (2013). Digital business strategy: Toward a next generation of insights. *MIS Quarterly*, 37(2), 471–482. <https://doi.org/10.25300/MISQ/2013/37.2.3>
- Bomm, E., De Montreuil Carmona, L. J., & Gomes, G. (2022). Unravelling t-KIBS performance: Leadership, organisational culture, and learning as boosters of service innovation. *Knowledge Management Research & Practice*, 1–14. <https://doi.org/10.1080/14778238.2022.2096510>
- Chen, J., Chen, Y., & Vanhaverbeke, W. (2011). The influence of scope, depth, and orientation of external technology sources on the innovative performance of Chinese firms. *Technovation*, 31(8), 362–373. <https://doi.org/10.1016/j.technovation.2011.03.002>
- Coronado-Medina, A., Arias-Pérez, J., & Perdomo-Charry, G. (2020). Fostering product innovation through digital transformation and absorptive capacity. *International Journal of Innovation and Technology Management*, 17(6), 1–23. <https://doi.org/10.1142/S0219877020500406>
- de Bem Machado, A., Secinaro, S., Calandra, D., & Lanzalonga, F. (2022). Knowledge management and digital transformation for Industry 4.0: A structured literature review. *Knowledge Management Research & Practice*, 20(2), 320–338. <https://doi.org/10.1080/14778238.2021.2015261>
- Duke, J., Igwe, V., Tapang, A., & Usang, O. (2022). The innovation interface between knowledge management and firm performance. *Knowledge Management Research & Practice*, 1–13. <https://doi.org/10.1080/14778238.2022.2029596>
- Erceg, V., & Zoranović, T. (2022). Knowledge management and digital business transformation. *Strategic Management*, 27(2), 57–63. <https://doi.org/10.5937/StraMan2200007E>
- Ferreira, J. J. M., Fernandes, C. I., & Ferreira, F. A. F. (2019). To be or not to be digital, that is the question: Firm innovation and performance. *Journal of Business Research*, 101, 583–590. <https://doi.org/10.1016/j.jbusres.2018.11.013>
- Forman, C., & Zeebroeck, N. V. (2019). Digital technology adoption and knowledge flows within firms: Can the internet overcome geographic and technological distance? *Research Policy*, 48(8), 103697. <https://doi.org/10.1016/j.respol.2018.10.021>
- Gao, H., Ding, X.-H., & Wu, S. (2020). Impact of knowledge search on product and process innovation: Mediating role of absorptive capacity and moderating role of IT capability. *European Journal of Innovation Management*, 25(2), 325–346. <https://doi.org/10.1108/EJIM-12-2019-0350>
- Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17(SI), 109–122. <https://doi.org/10.1002/smj.4250171110>
- Hassan, N., & Raziq, A. (2019). Effects of Knowledge Management Practices on Innovation in SMEs. *Management Science Letters*, 9(7), 997–1008. <https://doi.org/10.5267/j.msl.2019.4.005>
- Hautala-Kankaanpää, T. (2022). The impact of digitalization on firm performance: Examining the role of digital culture and the effect of supply chain capability. *Business Process Management Journal*, 28(8), 90–109. <https://doi.org/10.1108/BPMJ-03-2022-0122>
- Heberle, A., Lowe, W., Gustafsson, A., & Vorrei, O. (2017). Digitalization canvas- Towards identifying digitalization use cases and projects. *Journal of Universal Computer Science*, 23(11), 1070–1097. <https://doi.org/10.3217/jucs-023-11-1070>
- Hensen, A. H. R., & Dong, J. Q. (2020). Hierarchical business value of information technology: Toward a digital innovation value chain. *Information & Management*, 57(4), 103209. <https://doi.org/10.1016/j.im.2019.103209>
- Hu, P., Wang, Y., Feng, T., & Duan, Y. (2021). Innovative search, capability reconfiguration and firm innovation performance in the process of technological leapfrogging. *Chinese Management Studies*, 15(5), 961–984. <https://doi.org/10.1108/CMS-02-2020-0051>
- Jayakrishnan, M. A., Mohamad, A. K. B., & Yusof, M. B. M. (2018). Understanding holistic view and complexities in big data analytics and business intelligence (BI) towards establishing strategic performance management: A case study. *Advanced Science Letters*, 24(3), 1775–1779. <https://doi.org/10.1166/asl.2018.11157>
- Katila, R., & Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6), 1183–1194. <https://doi.org/10.5465/3069433>
- Lanzolla, G., Pesce, D., & Tucci, C. L. (2021). The digital transformation of search and recombination in the innovation function: Tensions and an integrative framework. *Journal of Product Innovation Management*, 38(1), 90–113. <https://doi.org/10.1111/jpim.12546>
- Laursen, K., & Salter, A. (2006). Open for innovation: The role of openness in explaining innovation performance among UK manufacturing firms. *Strategic Management Journal*, 27(2), 131–150. <https://doi.org/10.1002/smj.507>
- Lewin, A. Y., Massini, S., & Peeters, C. (2011). Microfoundations of internal and external absorptive capacity routines. *Organization Science*, 22(1), 81–98. <https://doi.org/10.1287/orsc.1100.052>
- Li, Y., Dai, J., & Cui, L. (2020). The impact of digital technologies on economic and environmental performance in the context of industry 4.0: A moderated mediation model. *International Journal of Production Economics*, 229, 107777. <https://doi.org/10.1016/j.ijpe.2020.107777>
- Martínez-Caro, E., Cegarra-Navarro, J. G., & Alfonso-Ruiz, F. J. (2020). Digital technologies and firm performance: The role of digital organisational culture. *Technological*

- Forecasting and Social Change*, 154, 119962. <https://doi.org/10.1016/j.techfore.2020.119962>
- Martins, E. C., & Terblanche, F. (2003). Building organisational culture that stimulates creativity and innovation. *European Journal of Innovation Management*, 6(1), 64–74. <https://doi.org/10.1108/14601060310456337>
- Matarazzo, M., Penco, L., Profumo, G., & Quaglia, R. (2021). Digital transformation and customer value creation in made in Italy SMEs: A dynamic capabilities perspective. *Journal of Business Research*, 123, 642–656. <https://doi.org/10.1016/j.jbusres.2020.10.033>
- McConnell, J. (2015). *The company cultures that help (or hinder) digital transformation*. *Harvard Business Review*, 2–6. <https://hbr.org/2015/08/the-company-cultures-that-help-or-hinder-digital-transformation>
- Mina, A., Bascavusoglu-Moreau, E., & Hughes, A. (2014). Open service innovation and the firm's search for external knowledge. *Research Policy*, 43(5), 853–866. <https://doi.org/10.1016/j.respol.2013.07.004>
- Nambisan, S. (2017). Digital entrepreneurship: Toward a digital technology perspective of entrepreneurship. *Entrepreneurship Theory and Practice*, 41(6), 1029–1055. <https://doi.org/10.1111/etap.12254>
- Nelson, R. R., & Winter, S. (1982). *An evolutionary theory of economic change*. Harvard University Press.
- Papa, A., Santoro, G., Tirabeni, L., & Monge, F. (2018). Social media as tool for facilitating knowledge creation and innovation in small and medium enterprises. *Baltic Journal of Management*, 13(3), 329–344. <https://doi.org/10.1108/BJM-04-2017-0125>
- Peng, Y., & Tao, C. (2022). Can digital transformation promote enterprise performance? —From the perspective of public policy and innovation. *Journal of Innovation & Knowledge*, 7(3), 100198. <https://doi.org/10.1016/j.jik.2022.100198>
- Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42(1), 185–227. <https://doi.org/10.1080/00273170701341316>
- Proksch, D., Rosin, A. F., Stubner, S., & Pinkwart, A. (2021). The influence of a digital strategy on the digitalization of new ventures: The mediating effect of digital capabilities and a digital culture. *Journal of Small Business Management*, 1–29. <https://doi.org/10.1080/00472778.2021.1883036>
- Roper, S., Love, J. H., & Bonner, K. (2017). Firms' knowledge search and local knowledge externalities in innovation performance. *Research Policy*, 46(1), 43–56. <https://doi.org/10.1016/j.respol.2016.10.004>
- Rosenkopf, L., & Nerkar, A. (2001). Beyond local search: Boundary spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4), 287–306. <https://doi.org/10.1002/smj.160>
- Scuotto, V., Caputo, F., Villasalero, M., & Del Giudice, M. (2017). A multiple buyer–supplier relationship in the context of SMEs' digital supply chain management. *Production Planning & Control*, 28(16), 1378–1388. <https://doi.org/10.1080/09537287.2019.1637135>
- Stuart, T. E., & Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(SI), 21–38. <https://doi.org/10.1002/smj.4250171004>
- Trantopoulos, K., von Krogh, G., Wallin, M. W., & Woerter, M. (2017). External knowledge and information technology: Implications for process innovation performance. *MIS Quarterly*, 41(1), 287–300. <https://doi.org/10.25300/MISQ/2017/41.1.15>
- Urbinati, A., Chiaroni, D., Chiesa, V., & Frattini, F. (2020). The role of digital technologies in open innovation processes: An exploratory multiple case study analysis. *R&D Management*, 50(1), 136–160. <https://doi.org/10.1111/radm.12313>
- Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research*, 122, 889–901. <https://doi.org/10.1016/j.jbusres.2019.09.022>
- Vial, G. (2019). Understanding digital transformation: A review and a research agenda. *The Journal of Strategic Information Systems*, 28(2), 118–144. <https://doi.org/10.1016/j.jsis.2019.01.003>
- Wamba, S. F., Gunasekaran, A., Akter, S., Ren, S. J., Dubey, R., & Childe, S. J. (2017). Big data analytics and firm performance: Effects of dynamic capabilities. *Journal of Business Research*, 70, 356–365. <https://doi.org/10.1016/j.jbusres.2016.08.009>
- Wang, T., Lin, X., & Sheng, F. (2022). Digital leadership and exploratory innovation: From the dual perspectives of strategic orientation and organizational culture. *Frontiers in Psychology*, 13, 902693. <https://doi.org/10.3389/fpsyg.2022.902693doi>
- Wu, L., Lou, B., & Hitt, L. (2019). Data analytics supports decentralized innovation. *Management Science*, 65(10), 4863–4877. <https://doi.org/10.1287/mnsc.2019.3344>
- Xia, H., Ye, P., Jasimuddin, S. M., & Zhang, J. Z. (2022). Evolution of digital transformation in traditional enterprises: Evidence from China. *Technology Analysis & Strategic Management*, 1–21. <https://doi.org/10.1080/09537325.2022.2124909>
- Yu, B., Hao, S., & Wang, Y. (2020). Organizational search and business model innovation: The moderating role of knowledge inertia. *Journal of Knowledge Management*, 24(7), 1705–1718. <https://doi.org/10.1108/JKM-02-2020-0100>
- Zhang, Y., & Li, H. (2010). Innovation search of new ventures in a technology cluster: The role of ties with service intermediaries. *Strategic Management Journal*, 31(1), 88–109. <https://doi.org/10.1002/smj.806>